

CONCEPT OF CURRENT AND ELECTRON FLOW.

The presence of free electrons enables current flow in a conducting material. The direction of current flow is always opposite to the electrons as shown in the figure below;



Fig (2) current flow is opposite to the electron.

ELECTRIC CURRENT (I): is simply defined as the rate of flow of charge, also is the time rate of charge passing through a predetermined area. Typically this area is the cross-sectional area of a metal wire. Consider a quantity charge Δq flowing through the cross-sectional area A in Δt units of time. The resulting current i is then given by;

$$i = \Delta q / \Delta t = c/s \quad \dots\dots\dots (1)$$

Considering the economic number of charges actually flowing, we can write this relationship in differential form;

$$i = \delta q / \delta t = c/s \quad \dots\dots\dots (2)$$

The unit of current are called ampere. Where 1 ampere (A) = 1 coulomb per second.

POTENTIAL DIFFERENCE (V): the voltage, or the potential difference between two point in a circuit indicates the energy require to move charge from one point to another, it's measure in J/C

$$\text{Potential difference (p.d) (V) = work done (J) / quantity of charge (C) } \dots\dots\dots (3)$$

$$\text{Therefore } 1 \text{ volt} = 1 \text{ joule (J) / coulomb (C) } \dots\dots\dots (4)$$

ELECTROMOTIVE FORCE (E.M.F): no resultant current will flow in a conductor unless there is an external source of energy creating an electrical force, in the conductor. The term e.m.f is used as a measure of the ability of a source to drive current around a circuit.

Formally the e.m.f is defined as work done in taking a unit charge completely round in a circuit. It is measures in volts.

RESISTANCE (R): is defined as the property of a substance which oppose (or resist) the flow of current through it or in a circuit. And it's measured in ohms (Ω).

MULTIPLES AND SUB-MULTIPLES OF ELECTRIC QUANTITIES.

Abbreviation	Prefix	Multiplier
T	Tera	10^{12}
G	Giga	10^9
M	Mega	10^6
K	Kilo	10^3
H	Hecto	10^2
M	Milli	10^{-3}
μ	Micro	10^{-6}
N	Nano	10^{-9}
P	Pico	10^{-12}

Table (1) multiples and sub-multiples of electrical quantity.

THE ELECTRIC CIRCUIT

An electric circuit is the complete path provided for the flow of electric current. The circuit usually consists of the source of electric energy; the battery (or cell), a switch or key to start or stop the current flow, an ammeter used to measure the flow of current, a voltmeter used to measure the potential difference, a resistor or load and a rheostat used to adjust the flow of current in the circuit.

Volt meter

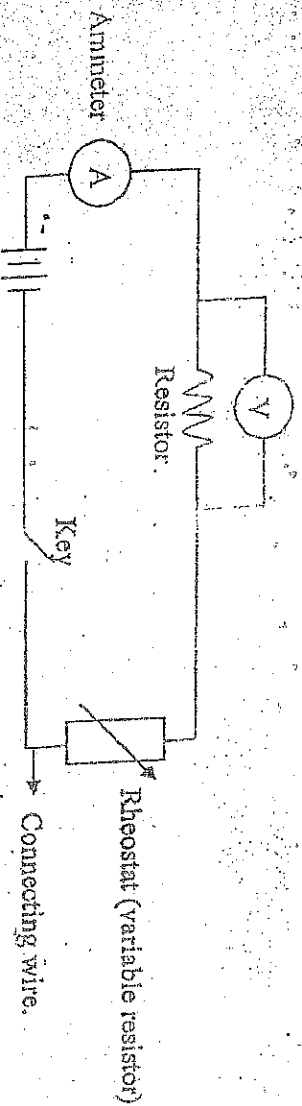


Fig (3) Basic d.c circuit.

Direct current (d.c); direct current always flows in the same direction, but it may increase or decrease. d.c current can be defined as an electric current which flows consistently in one direction only or is the flow of charge in one direction.

ANALOGY BETWEEN CURRENT FLOW AND WATER FLOW.

As stated earlier, potential difference is an essential condition for the flow of current in an electric circuit. We can get a feel of its effect on current flow by considering the flow of water along a pipe. For the water to flow along a pipe there must be a pressure difference between one end of the pipe and the other. Similarly for electric current to flow along a conductor, there must be electrical pressure difference between the ends of the conductor.

OHM'S LAW:

This law applies to electric conduction through good conductors and may be stated as the current passing through a metallic conductor is directly proportional to the potential difference across it's terminals or ends, provided that the temperature and other physical conditions of the conductors remain or kept constant.

In symbols, ohms law states that

$$I \propto V \dots\dots\dots (5a)$$

Removing the proportionality sign we have that

$$V/I = \text{constant or } V/I = R \dots\dots\dots (5b)$$

Where R is the resistance of the conductor between the two points consider,

$$\text{Mathematically, } V = IR \dots\dots\dots (5c)$$

$$I = V/R \dots\dots\dots (5d)$$

EXPERIMENTAL VERIFICATION OF OHM'S LAW: we can verify ohms law using the set up of apparatus as shown below,

Voltmeter

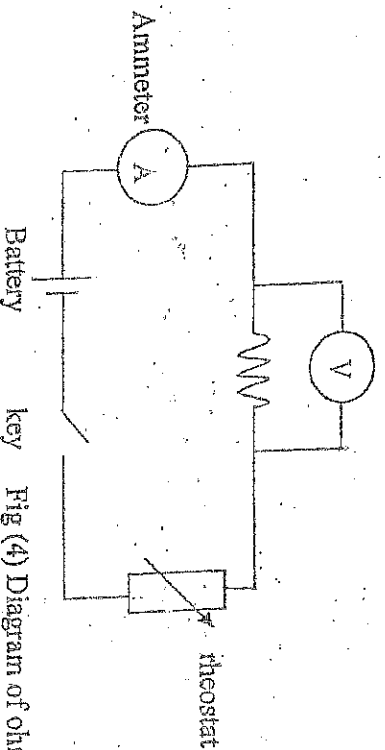
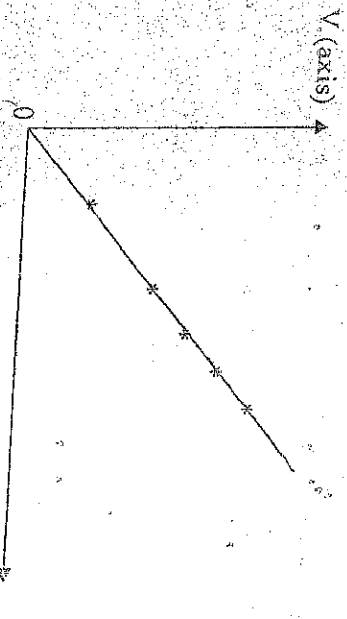


Fig (4) Diagram of ohms law.

We cause current to flow in the circuit by pressing the key. The value of current (I) and p.d (V) recorded by both the ammeter and voltmeter respectively are recorded. The rheostat is adjusted to a new setting, and the new values of I and V are read and recorded. This process is repeated for more times to obtain multiple results, which are now set up in a table.

The graph of V against I is plotted and a straight line graph passing through the origin is obtained. The straight line graph shows that V varies directly with I thus verifying ohm's law. The slope of the graph gives the resistance of the wire. The graph of V against I is shown in figure below.



$$= \frac{V}{I} = 1$$

$$V = I$$

$$R = 1$$

Fig (5) the graph of V against I (graph of ohm's law).

Example (1); complete the following table assuming that the pieces of apparatus follow ohm's law.

Apparatus	Current (A)	p.d (V)	Resistance (R)
Voltmeter	0.50	5.67	600
Ammeter	5	0.06	0.3
Power line	40	1.0	4
Torch bulb	0.25	2	8

RESISTIVITY AND CONDUCTIVITY OF A CONDUCTOR

The resistance R offered by a conductor depends on the following factors;

- (1) It varies directly as its length, L
 $R \propto L$ (6.a)
- (2) It varies inversely as the cross-sectional area A of the conductor.
 $R \propto 1/A$ (6.b)
- (3) It depends on the nature of the material.
- (4) It also depends on the temperature of the conductor.

Neglecting the last factor for the time being, we can say that

$$R \propto L/A \text{ or } R = \rho L/A$$

Where ρ is the constant depending on the nature of the material of the conductor and is known as its specific resistance or resistivity. (6.c)

RESISTIVITY (ρ); can be defined as the resistance between the opposite faces of a meter cube of that material or resistivity is the measure of resistance in that material.
 from equation (6.c) $\rho = AR/L$ (Ohm) (6.d)

CONDUCTANCE (G); is the reciprocal of resistance. Whereas resistance of a conductor measures the opposition which it offers to the flow of current, the conductance measures the inducement which it offers to its flow. Its measures in Siemens (s). Earlier this unit is mho.

$$\text{From equation (6.c), } R = \rho L/A$$

$$\text{But } G = 1/R = 1/\rho \cdot A/L = GA/L$$

Therefore $G = GA/L$ (s) (7.a)

Where G is called the conductivity or specific conductance of a conductor (7.b)

CONDUCTIVITY (σ), is a measure of the extent to which a material will allow current to flow easily through it when a p.d is applied at a specified temperature. It's the reciprocal of the resistivity. It's as seen in equation (7.b) that conductivity of a material is given by;

$$\sigma = GL / A \quad (\text{S/m})$$

Example (2);

(a) The dimensions of a rectangular carbon block are 2.0cm by 2.0cm by 50cm. what is

- (i) The resistance measured between the two square ends?
- (ii) The resistance between two opposite rectangular faces. The resistivity of carbon at 20°C is $3.5 \times 10^{-5} \Omega \text{m}$.

(b) A current of 5A exist in a 10Ω resistance for 4 minute.

- (i) How many coulombs and
- (ii) How many electrons pass through any section of the conductor in this time? Electron charges is $1.6 \times 10^{-19} \text{C}$.

Solution;

(a) (i) using: $R = \rho L / A$

Therefore; Here $A = 2\text{cm} \times 2\text{cm} = 4\text{cm}^2 = 4 \times 10^{-4} \text{m}^2$, $L = 50\text{cm} = 0.5\text{m}$
 $R = 3.5 \times 10^{-5} \times 0.5 / 4 \times 10^{-4} \text{m}^2$
 $R = 0.044\Omega$

Therefore; (ii) Here $A = 2\text{cm} \times 50\text{cm} = 100\text{cm}^2 = 100 \times 10^{-4} \text{m}^2$, $L = 2\text{cm} = 2 \times 10^{-2} \text{m}$
 $R = 3.5 \times 10^{-5} \times 2 \times 10^{-2} / 100 \times 10^{-4} \text{m}^2$
 $R = 7 \times 10^{-4} \Omega$

- (b) (i) Using $Q = It = 5 \times (4 \times 60) = 1200\text{C}$.
- (ii) $n = Q / e = 1200 / 1.6 \times 10^{-19} = 7.5 \times 10^{20}$

Example (3);

The conductivity of a ferric-chromium-aluminium alloy is $2 \times 10^6 \text{ S/m}$. A sheet of the material is 15cm long, 6cm wide and 0.014cm thick. Determine the conductance between the opposite sides.

Solution; $L = 0.014\text{cm} = 14 \times 10^{-5} \text{m}$, and $A = 15 \times 6 = 90\text{cm}^2 = 90 \times 10^{-4} \text{m}^2$
 Therefore $G = \sigma A / L = 2 \times 10^6 \times 90 \times 10^{-4} / 14 \times 10^{-5} = 1.3 \times 10^8 \text{S}$
 $G = 1.3 \times 10^8 \text{S}$

SERIES AND PARALLEL CIRCUIT

When looking at the series and parallel circuits, there is a certain peculiar pattern of relationship between them. For example, in a series circuit, current is the same where as in parallel circuit, voltage is the same. Also in a series circuit, individual voltages are added and in parallel circuit, individual current are added. It's seen that when comparing series and parallel circuits, voltage takes the place of current and current takes the place of voltage. Such a pattern is known as duality, and the circuit are said to be duals of each other.

RESISTANCE IN SERIES.

Consider a conductor having resistances R_1, R_2, R_3 etc joined end on end as shown in fig (6), below they are said to be in series. It can be proved that the equivalent resistance or total resistance between points A and D is equal to the sum of those individual resistances.

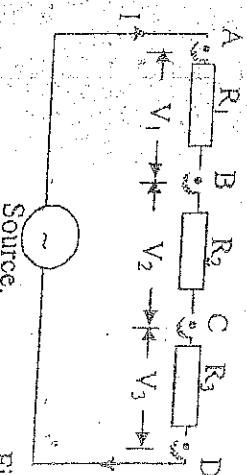


Fig (6) Series Arrangement of Resistors.

Therefore $V = V_1 + V_2 + V_3 = IR_1 + IR_2 + IR_3$
 But $V = IR$

Where R is the equivalent resistance of the series combination.

$$IR = IR_1 + IR_2 + IR_3$$

$$R = R_1 + R_2 + R_3$$

Also $1/G = 1/G_1 + 1/G_2 + 1/G_3$

As seen from above, the main characteristics of series circuits are;

- (1) Same current flow through all parts of the circuit.
- (2) Difference resistors have their individual voltage drop.
- (3) Voltage drop are additive.
- (4) Applied voltage equal to the sum of different voltage drop.
- (5) Resistance are additive.
- (6) Powers are additive.

RESISTANCE IN PARALLEL:

The three resistors as joined in fig (7) are said to be connected in parallel. In this case the p.d across all resistors is the same, current in each resistor is different and the total current is the sum of the three separate current passing through the resistors.

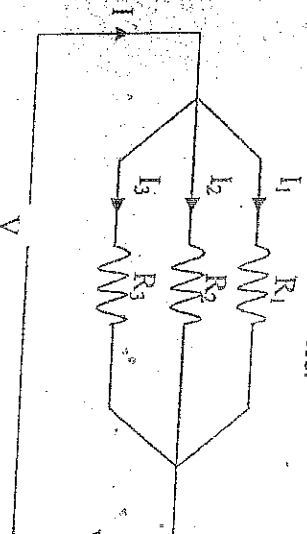


Fig (7) Parallel Arrangement of Resistors

Therefore $I = I_1 + I_2 + I_3 = V/R_1 + V/R_2 + V/R_3$
 But $I = V/R$, where R is the equivalent parallel resistance.

Also $V/R = V/R_1 + V/R_2 + V/R_3$ or $1/R = 1/R_1 + 1/R_2 + 1/R_3$
 Also $G = G_1 + G_2 + G_3$

The main characteristics of a parallel circuit are;

- (1) Same voltage acts across all parts of the circuit.
- (2) Different resistors have their individual current.
- (3) Branch currents are additive.
- (4) Conductance are additive.
- (5) Powers are additive.

VOLTAGE DIVISION IN SERIES CIRCUIT (VOLTAGE DIVIDER RULE):

Since in a series circuit, same current flows through each of the given resistors, voltage drop varies directly with its resistance.

In fig (8) shown below a 24V battery connected across a series combination of three resistors.

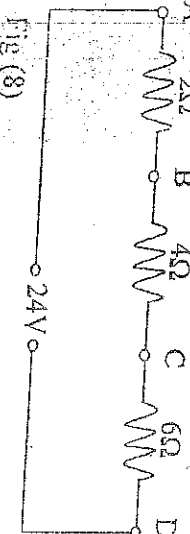


Fig (8)

Total resistance $R_T = R_1 + R_2 + R_3 = 2 + 4 + 6 = 12\Omega$
 According to voltage divider rule, various voltage drops across are;

$$V_1 = V \times R_1 / R = 24 \times 2 / 12 = 4V$$

$$V_2 = V \times R_2 / R = 24 \times 4 / 12 = 8V$$

$$V_3 = V \times R_3 / R = 24 \times 6 / 12 = 12V.$$

CURRENT DIVISION IN PARALLEL CIRCUIT (CURRENT DIVIDER RULE): In the case of three resistors in parallel as shown in fig (9),

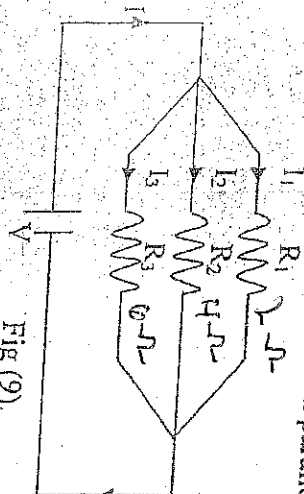


Fig (9).

Total current $I = I_1 + I_2 + I_3$, and let R_T = total resistance.
 Then $V = IR$.

Also $V = I_1 R_1$

therefore $IR = I_1 R_1 \implies I_1 = IR / R_1$

Now $I / R = 1 / R_1 + 1 / R_2 + 1 / R_3$

$$R = R_1 R_2 R_3 / R_2 R_3 + R_1 R_3 + R_1 R_2$$

$$I_1 = I / R_1 \{ R_1 R_2 R_3 / R_2 R_3 + R_1 R_3 + R_1 R_2 \} = I \{ R_2 R_3 / R_2 R_3 + R_1 R_3 + R_1 R_2 \}$$

$$I_1 = I \{ R_2 R_3 / R_2 R_3 + R_1 R_3 + R_1 R_2 \}$$

Similarly $I_2 = I \{ R_1 R_3 / R_2 R_3 + R_1 R_3 + R_1 R_2 \}$

$$I_3 = I \{ R_1 R_2 / R_2 R_3 + R_1 R_3 + R_1 R_2 \}$$

KIRCHOFF'S LAW:

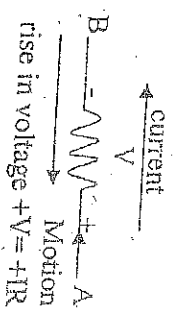
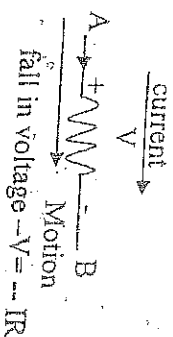
(1). Kirchhoff's voltage law (KVL);

Kirchoff's voltage law states that the algebraic sum of the products of current and resistances in each of the conductor in any closed path (or mesh) in a network plus the algebraic sum of the e.m.f in that path is zero.

In other words, $\sum IR + \sum e.m.f = 0$.

It should be noted that algebraic sum is the sum which takes into account the polarities of the voltage drops.

Sign of battery E.M.F



It's important to note that the sign of the battery e.m.f is independent of the direction of current through that branch.

$$\begin{aligned} I_1 R_1 &= -V_e \\ I_2 R_2 &= -V_e \\ I_3 R_3 &= +V_e \end{aligned}$$

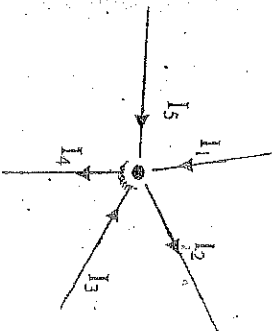
$$\begin{aligned} I_4 R_4 &= -V_e \\ E_1 &= +V_e \\ E_2 &= -V_e \end{aligned}$$

Applying kirchoffs voltage law to the mesh ABCDA.
 $-I_1 R_1 - I_2 R_2 + I_3 R_3 - I_4 R_4 - E_2 + E_1 = 0$
 or $I_1 R_1 + I_2 R_2 - I_3 R_3 + I_4 R_4 = E_1 - E_2$

(2) Kirchoff's current law (KCL);

Kirchoff's current law states that in any electrical network, the algebraic sum of the current meeting at a point (or junction) is zero. Or simply means that the total current entering a junction is equal to the total current entering that junction.

Consider the case of a few conductors meeting at a point A in fig (11). Some conductors have current leading to point A, whereas some have currents leading away from point A.



Fig(11)

therefore $\Sigma I = 0$

Assuming the incoming currents to be positive and the outgoing current negative.
 $I_1 + (-I_2) + I_3 + (-I_4) + I_5 = 0$

$I_1 - I_2 + I_3 - I_4 + I_5 = 0$ or $I_1 + I_3 + I_5 = I_2 + I_4$
 Or incoming current = outgoing current.

NETWORK ANALYSIS USING KIRCHOFF'S LAWS

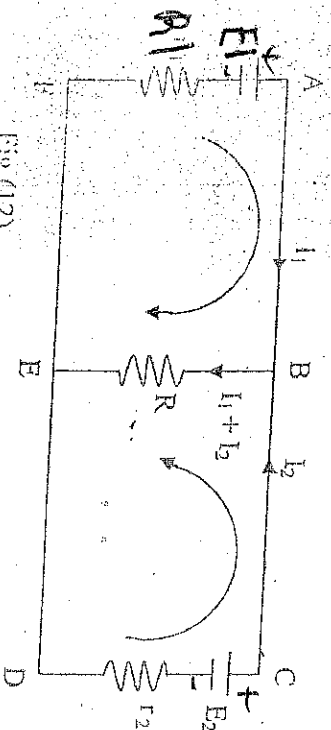


Fig (12)

- To calculate the current flowing in each branch, the following three steps or procedure may be used.
- Label the branch current and their directions on the circuit diagram.
 - Divide the circuit into loops an apply KVL to each loop intum.

$$R(I_1 + I_2) + I_1 r_1 - E_1 = 0$$

$$-E_1 = I_1 (r_1 + R) + I_2 R$$

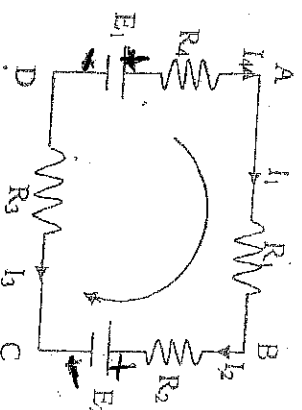
From loop BEDCB and moving anti-clockwise;.....(8.a)

$$R(I_1 + I_2) + I_2 r_2 - E_2 = 0$$

$$E_2 = I_2 (r_2 + R) + I_1 R$$

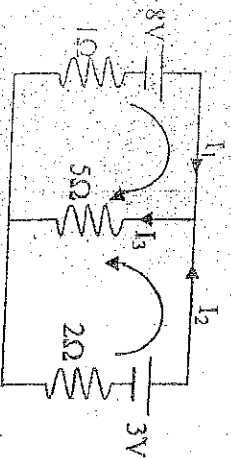
- Solve (8.a) and (8.b) simultaneously or otherwise.(8.b)

Example (4): find I_1 and I_2 ? If $E_1 = 8V$, $E_2 = 3V$, $r_1 = 1\Omega$, $r_2 = 2\Omega$, $R = 5\Omega$.



Fig(10)

Solution:



Substituting the above values in equations (8.a) and (8.b)

$$I_3 = I_1 + I_2$$

$$6I_1 + 5I_2 = 8$$

$$5I_1 + 7I_2 = 3$$

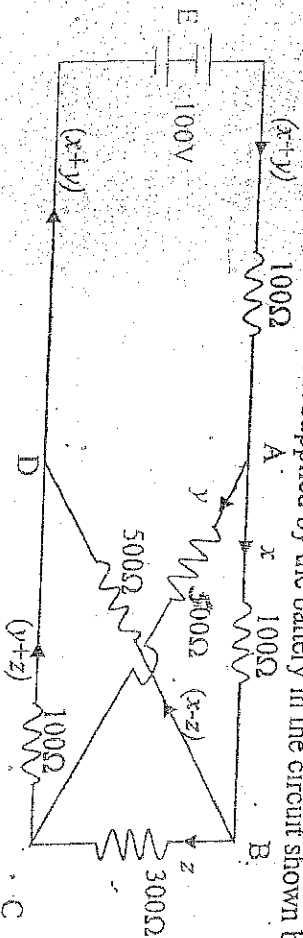
Using determinant method we have;

$$I_1 = \frac{1}{\begin{vmatrix} 6 & 5 \\ 5 & 7 \end{vmatrix}} \begin{vmatrix} 8 & 3 \\ 3 & 5 \end{vmatrix} = \frac{41}{17} = 2.41A$$

$$I_2 = \frac{1}{\begin{vmatrix} 6 & 5 \\ 5 & 7 \end{vmatrix}} \begin{vmatrix} 6 & 8 \\ 5 & 3 \end{vmatrix} = \frac{-22}{17} = -1.29A$$

(Confirms the actual direction of the flow of current)
 $I_2 = -1.29A$. (Indicate that the actual direction is opposite to that assumed)

Example (5); determine the current supplied by the battery in the circuit shown below



Solution: let the distribution of current to be as shown in the figure above. Applying KVL to ABCEA
 $100x - 300x + 500y = 0$ multiply this equation by 1 / 100.
 $x + 5y + 3z = 0$ (9.a)

Similarly considering closed loop BCDB, we have;
 $300z - 100(x + z) + 500(x - z) = 0$
 $3x - y - 9z = 0$ (9.b)

Also taking the loop ABDEA, we get
 $-100x - 500(x - z) + 100 - 100(x + y) = 0$
 $7x + y - 5z = 1$ (9.c)

The value of x , y , and z may be found by solving the three equations (9.a), (9.b) and (9.c) above gives;

$$\begin{vmatrix} 1 & 5 & 3 \\ 5 & -1 & -9 \\ 7 & 1 & -5 \end{vmatrix} \begin{vmatrix} x \\ y \\ z \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \\ 1 \end{vmatrix}$$

$$\Delta = \begin{vmatrix} 1 & 5 & 3 \\ 5 & -1 & -9 \\ 7 & 1 & -5 \end{vmatrix} = -240$$

$$\Delta_1 = \begin{vmatrix} 0 & 5 & 3 \\ 0 & -1 & -9 \\ 1 & 1 & -5 \end{vmatrix} = -42$$

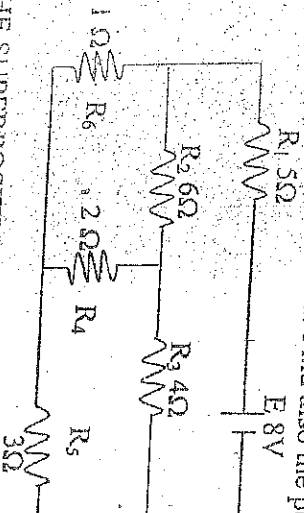
$$\Delta_2 = \begin{vmatrix} 1 & 0 & 3 \\ 5 & 0 & -9 \\ 7 & -1 & -5 \end{vmatrix} = 24$$

$$\Delta_3 = \begin{vmatrix} 1 & 5 & 0 \\ 5 & -1 & 0 \\ 7 & 1 & 1 \end{vmatrix} = -26$$

Therefore $x = \Delta_1 / \Delta = 48 / 240 = 1 / 5A$.
 $y = \Delta_2 / \Delta = 24 / 240 = 1 / 10A$.
 $z = \Delta_3 / \Delta = 24 / 240 = 1 / 10A$.

Therefore: the current supply by the battery is $x + y = 1 / 5 + 1 / 10 = 3 / 10A$.

Exercise (1): determine the current flowing in the 2Ω resistor of the circuit shown below using Kirchhoff's law and determinant method. Find also the power dissipated in the 3Ω resistance.



THE SUPERPOSITION THEORY

"In any network made up of linear impedances and containing more than one source of e.m.f the resultant current flowing in any branch is the phasor sum of the current that would flow in that branch if each source were considered separately and all other sources being replaced at that time by their respective internal resistances".

The superposition theorem may be applied to d.c and a.c network.

Example (6): use superposition theorem to analyzed fig (13) a.

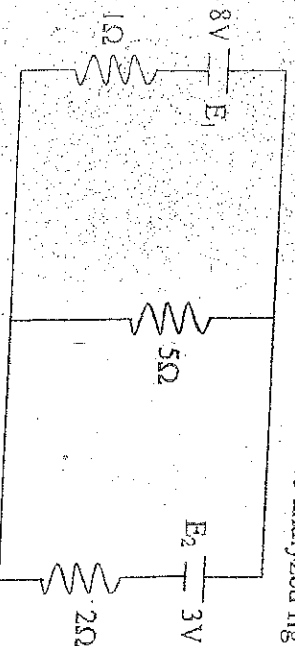


Fig (13.a)

Solution:

The following steps can be used to analyze the circuit.
 Step (1): Redraw the original network with E_2 short circuited.
 Step (2): Label the current in each branch and if direction and determine its value (conventional direction of current must be followed for sources).

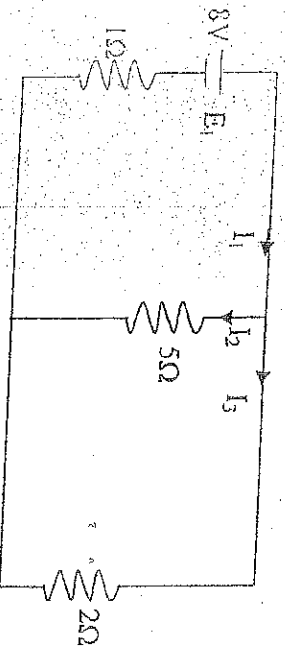


Fig (13.b).

Thus total resistance $R_{T1} = \frac{5 \times 2}{5 + 2} + 1 = 2.429\Omega$

Therefore $I_1 = 8 / 2.429 = 3.294A$

From fig (13.b): $I_2 = 3.294 \times 2 / 7 = 0.941A$

$I_3 = 3.294 \times 5 / 7 = 2.353A$

Step (3): Redraw the original network again with E_1 short circuited.

Step (4): Label the current in each branch and find its value as well.

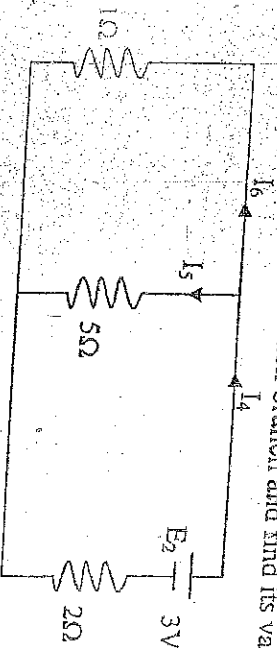


Fig (13.c)

As before,

$$R_{T2} = \frac{1 \times 5}{1 + 5} + 2 = 2.833\Omega$$

Therefore $I_4 = 3 / 2.833 = 1.059A$

From fig (13.c); $I_5 = 1.059 \times 1 / 6 = 0.177A$

$$I_6 = 1.059 \times 5 / 6 = 0.883A.$$

Step (5): superimpose fig (13.b) and (13.c) as shown in fig (13.d),

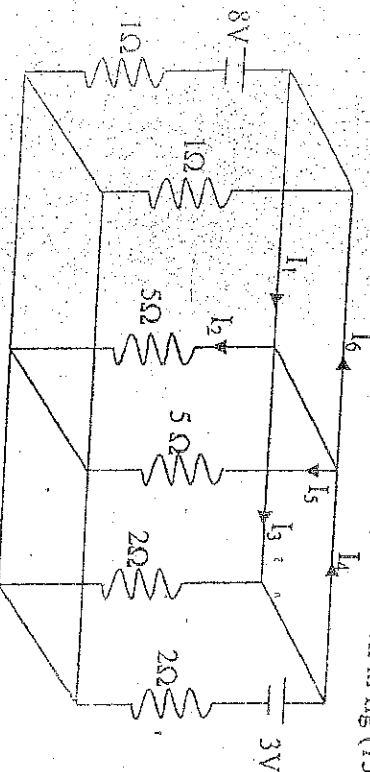


Fig (13.d)

Step (6): determines the algebraic sum of the current flowing in each branch from fig (13.d).
 The resultant current through the 8V source is:

$$I_1 - I_6 = 3.294 - 0.883 = 2.41A \text{ (discharging)}$$

The resultant current through the 3V source is

$$I_4 - I_3 = 1.059 - 2.353 = -1.29A \text{ (charging)}$$

And the resultant current through the 5Ω resistor is;
 $I_5 + I_5 = 0.941 + 0.177 = 1.12A.$

Example (7) use superposition theorem to find the current in the 10Ω resistor of fig (14.a)

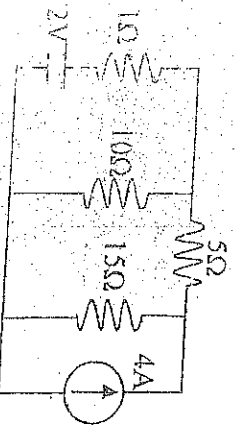


Fig (14.a)

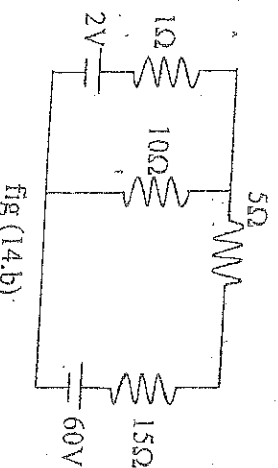
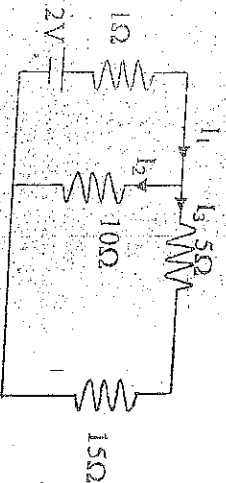


fig (14.b)

Solution;

The 4A source and its parallel resistance can be converted to a voltage source of $(15 \times 4) = 60V$ in series with 15Ω resistor as shown in fig (14.b).
 When 60V source is removed;

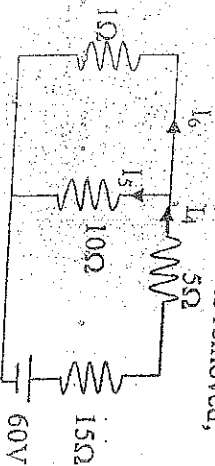


$$R_{T1} = 1 + \{ 20 \times 10 / 30 \} = 7.67\Omega.$$

$$I_1 = 2 / 7.67 = 0.26A$$

$$I_2 = 0.26 \times 20 / 30 = 0.17A$$

When 2V source is removed;



$$R_{T2} = 20 + \{ 1 \times 10 / 11 \} = 20.9\Omega$$

$$I_4 = 60 / 20.9 = 2.87A$$

$$I_5 = 2.87 \times 1 / \{ 10 + 1 \} = 0.26A$$

Total current in the 10Ω resistor = $I_2 + I_5 = 0.17 + 0.26 = 0.43A$
 Therefore $I_T = 0.43A$

TEMPERATURE COEFFICIENT OF RESISTANCE:

Consider a metallic conductor having a resistance of R_0 at $0^\circ C$ and let it be heated to $t^\circ C$ and let its resistance at this temperature be R_t then, considering normal ranges of temperature, it is found that the increase in resistance is $\Delta R = R_t - R_0$ depends on,

- (i) Directly on its initial resistance.
- (ii) Directly on the rise in temperature.
- (iii) On the nature of the material of the conductor.

or $R_t - R_0 \propto R_0 \times t$ or $R_t - R_0 = \alpha R_0 t$ (10.a)

Hence the temperature coefficient of a material may be defined as the increase in resistance per ohm original resistance per $^\circ C$ rise in temperature.

Rearranging equation (10.a) above we get

$$\text{If } R_0 = 1\Omega, t = 1^\circ C \text{ then } \alpha = \Delta R = R_t - R_0, \quad \alpha = R_t - R_0 / R_0 \times t = \Delta R / R_0 \times t.$$

From equation (10.a) we find that $R_t = R_0 (1 + \alpha t)$ (10.b)

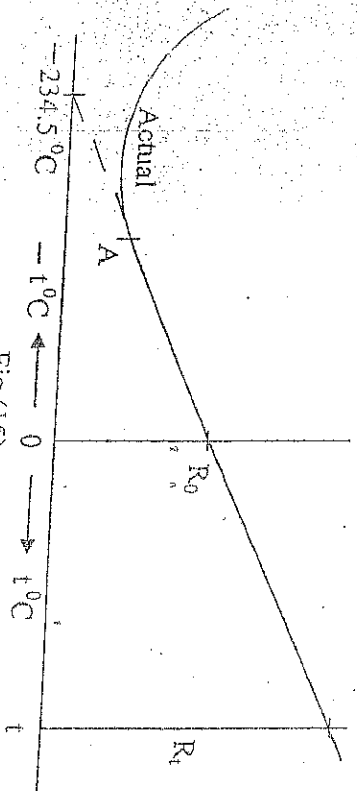


Fig (15)

It should be remember that the above equation holds good for both rise as well as fall in temperature. As temperature of a conductor is decreased, its resistance also decreased. In fig (15) is shown that the temperature / resistance graph for copper is practically a straight line graph. If this line is extended backward, it would cut the temperature axis at a point where temperature is at $-234.5^\circ C$. It means that theoretically that the resistance of copper conductor will become zero at this point, though as shown by solid line, in practice, the curve depart from a straight line at very low temperature.

From fig (15) it seen that

$$R_t / R_0 = 1 + 234.5 / 234.5 = (1 + t / 234.5)$$

$$R_t = R_0 (1 + t / 234.5) \quad \text{or} \quad R_t = R_0 (1 + \alpha t)$$

Where $\alpha = 1 / 234.5$ for copper.

VALUE OF α AT DIFFERENT TEMPERATURE

It is found that the value of α itself is not constant but depends on the initial temperature on which the increment in resistance is based. When the increment is based on the resistance measured at 0°C then α has the value of α_0 . At any other initial temperature $t_1^\circ\text{C}$, value of α is α_1 and so on. It should be remembered that, for any conductor, α_0 has the maximum value.

Suppose a conductor of resistance R_0 at 0°C is heated to $t^\circ\text{C}$. Its resistance R_t after heating is given by,

$$R_t = R_0 (1 + \alpha_0 t) \quad \dots\dots\dots (11.a)$$

Where α_0 is the temperature coefficient at 0°C .

Suppose that we have a conductor of resistance R_t at temperature $t^\circ\text{C}$. Let this conductor be cooled from $t^\circ\text{C}$ to 0°C . Obviously the final resistance R_0 is given in terms of the initial resistance by the following equation

$$R_0 = R_t [1 + \alpha_1 (-t)] = R_t [1 - \alpha_1 \times t] \quad \dots\dots\dots (11.b)$$

From equation (11.b) above, we have

$$\alpha_1 = R_t - R_0 / R_t \times t$$

Substituting the value of R_t from equation (11.a), we get

$$\alpha_1 = R_0 (1 + \alpha_0 t) - R_0 / R_0 (1 + \alpha_0 t) \times t = \alpha_0 / 1 + \alpha_0 t$$

Therefore,

$$\alpha_1 = \alpha_0 / 1 + \alpha_0 t \quad \dots\dots\dots (11.c)$$

In general let α_1 = temperature coefficient at $t_1^\circ\text{C}$;

α_2 = temperature coefficient at $t_2^\circ\text{C}$, then from equation (11.c) above, we get

$$\alpha_1 = \alpha_0 / 1 + \alpha_0 t_1 \quad \text{or} \quad 1 / \alpha_1 = 1 + \alpha_0 t_1 / \alpha_0 \quad \dots\dots\dots (11.d)$$

$$1 / \alpha_2 = 1 + \alpha_0 t_2 / \alpha_0 \quad \dots\dots\dots (11.e)$$

Substituting equation (11.d) from (11.e) we have

$$1 / \alpha_2 - 1 / \alpha_1 = (t_2 - t_1) \quad \text{or}$$

$$1 / \alpha_2 = 1 / \alpha_1 + (t_2 - t_1) \quad \text{or}$$

$$\alpha_2 = [1 / \alpha_1 + (t_2 - t_1)]^{-1}$$

In case R_0 is not given the relationship between the known resistance R_1 at $t_1^\circ\text{C}$ and the unknown resistance R_2 at $t_2^\circ\text{C}$ can be found as follows.

$$R_2 = R_0 (1 + \alpha_0 t_2) \quad \text{and} \quad R_1 = (1 + \alpha_0 t_1) R_0$$

$$R_2 / R_1 = [1 + \alpha_0 t_2] / [1 + \alpha_0 t_1] \quad \dots\dots\dots (11.f)$$

The above expression can be simplified by a little approximation as follows.

$$R_2 / R_1 = (1 + \alpha_0 t_2) (1 + \alpha_0 t_1)^{-1}$$

$$R_2 / R_1 = (1 + \alpha_0 t_2) (1 - \alpha_0 t_1)$$

Using binomial theorem for expansion and neglecting square and higher powers of $(\alpha_0 t_1)$ neglecting product $(\alpha_0^2 t_1 t_2)$

$$= 1 + \alpha_0 (t_2 - t_1)$$

$$\text{Therefore } R_2 = R_1 [1 + \alpha_0 (t_2 - t_1)]$$

But for more accuracy in calculation, equation (11.f) should be used. (11.g)

VARIATION OF RESISTIVITY WITH TEMPERATURE:

If P_1 = resistivity at t_1 °C

P_2 = resistivity at t_2 °C

Then $P_2 = P_1 [1 + \alpha_1 (t_2 - t_1)]$ or simply as $P_1 = P_0 (1 + \alpha_0 t)$.

Example (8): a platinum coil has a resistance of 3.146Ω at 40°C and 3.767Ω at 100°C. Find the resistance at 0°C and the temperature – coefficient of resistance at 40°C.

Solution:

$$R_{40} = R_0 (1 + 40\alpha_0)$$

$$R_{100} = R_0 (1 + 100\alpha_0).$$

Therefore:

$$3.767 / 3.146 = \cancel{R_0} (1 + 100\alpha_0) / \cancel{R_0} (1 + 40\alpha_0) = [1 + 100\alpha_0 / 1 + 40\alpha_0].$$

$$\alpha_0 = 0.00379 \text{ or } 1 / 264 \text{ per } ^\circ\text{C}.$$

Resistance at 0°C can be found as follows;

$$R_{40} = R_0 (1 + 40\alpha_0)$$

$$3.146 = R_0 (1 + 40 \times 0.00379)$$

$$R_0 = 3.146 / 1.1516 = 2.732.$$

$$R_0 = 2.732\Omega.$$

$$R_t = R_0 (1 + \alpha_0 t)$$

Temperature coefficient of resistance at 40°C;

$$\alpha_{40} = \alpha_0 / 1 + 40\alpha_0 = 0.00379 / 1 + 40 \times 0.00379 = 0.0033 / ^\circ\text{C}.$$

$$\alpha_{40} = 0.0033 / ^\circ\text{C} \text{ or } 1 / 303 / ^\circ\text{C}.$$

Example (9): a potential difference of 250V is applied to a field winding at 15°C and the current is 5A. what will be the mean temperature of the winding when the current falls to 3.91A, applied voltage being constant. Assume $\alpha_{15} = 1 / 254.5$.

Solution:

Let R_1 = winding resistance at 15°C;

R_2 = winding resistance at unknown mean temperature t_2 °C;

Therefore $R_1 = 250 / 5 = 50\Omega$. and $R_2 = 250 / 3.91 = 63.94\Omega$.

Now $R_2 = R_1 [1 + \alpha_{15} (t_2 - t_1)]$.

$$63.94 = 50 [1 + 1 / 254.5 (t_2 - 15)].$$

$$\text{Therefore } t_2 = 86^\circ\text{C}.$$

EXERCISE (2): a coil of copper wire has resistance of 90Ω at 20°C and is connected to a 230V supply. By how much must the voltage be increased in order to maintain the current constant if the temperature of the coil rises to 60°C? Take the temperature coefficient of resistance of copper as 0.0042 from 0°C.

ENERGY

There are many forms of energy. Examples are;

- (a) Mechanical energy which is made up of potential and kinetic energy.
- (b) Heat energy; used by steam engine to do work.
- (c) Light energy
- (d) Sound energy
- (e) Electrical energy
- (f) Nuclear energy; the energy in the nucleus of atoms produces heat energy which can be used to generate electrical energy.

ENERGY RESOURCES:

There are two main groups.

- (i) Renewable resources which can be used over and over again. And
 - (ii) Non renewable resources which can be used only once.
- Examples of renewable resources are;

- > Solar
- > Wind
- > Hydro
- > Ocean waves and tides.

Examples of non-renewable resources are;

- > Petroleum
- > Coal
- > Nuclear energy
- > Human activities.

RELATIONSHIP BETWEEN ELECTRICAL MECHANICAL AND THERMAL ENERGY.

One form of energy can readily be converted into another with the help of a device. For instance electrical energy is converted into mechanical energy by an electric motor, and mechanical energy is converted into electrical energy by a generator. Thermal energy (heat energy) is converted into mechanical energy by a steam turbine, and electrical energy is converted into thermal energy by a resistor. There is always a relationship between electrical, mechanical and thermal energy. Almost all electrical systems tend to exhibit either of the two forms of energy.

JOULE'S LAW OF ELECTRIC HEATING.

Joule was the first person to investigate the heating effect of an electric current, known as joules laws of electrical heating.

The heat energy H developed in a wire is proportional to;

- (a) The time t (for a given resistance and current)
- (b) The square of the current I^2 (for a given resistance and time).
- (c) The resistance R of the wire (for a given current and time).

Therefore the energy or heat produce is given by;

$$H = I^2 R t \text{ (J)}$$

Using ohms law;

$$H = V^2 t / R \text{ (J)}. \text{ Or } H = V I t \text{ (J)}$$

This heat energy is also the amount of work required to maintain a current I amperes through a resistance of R ohms for time t .

$$\text{Therefore } W.D = I^2 R t = H.$$

Example (10). Find the heat in joules produced in an electric coil of resistance 15Ω when a current of $5A$ flows through it for 30 minutes.

Solution, Given $R = 15\Omega$, $I = 5A$, $t = 30 \text{ minutes} = 30 \times 60s = 1800s$ ✓
Therefore using, $H = I^2 R t$ ~~$H = I^2 R t$~~

$$H = (5)^2 \times 15 \times 30 \times 60s.$$

$$H = 675000J \text{ or } 675KJ.$$

Example (11). A bulb takes a current of $5A$ from a $220V$ supply. What is the heat produced in 5 minutes?

Solution; Given $I = 5A$, $V = 220V$ and $t = 5 \text{ minutes} = 5 \times 60s$.

$$\text{Therefore } H = V I t$$

$$H = 220 \times 5 \times 5 \times 60$$

$$H = 330000J \text{ or } 330KJ.$$

Example (12). An electric lamp is labeled $25V$ at $120W$ atts.

- (i) What is the energy the lamp will dissipate per minutes when correctly used?
 - (ii) What is the rating in the fuse that will be adequate for the lamp?
- Solution;

- (i) Energy = $V I t$, but power = $V I = 120W$, and time = 1 minute = $60s$.

0071
750
23

Therefore energy = $120 \times 60 = 7200\text{J}$.

(ii) $V_1 = 120\text{V}$ and $V = 25\text{V}$.

Therefore $I = 120\text{V} / 25\text{V} = 4.8\text{A}$.

That is 4.8A is the rating of the fuse adequate for the lamp.

ELECTRIC CHARGE/ELECTROSTATIC:

In the previous lesson, we concern ourselves exclusively with electric current that is electricity in motion. Now we will discuss the behaviors of static electricity and the laws governing it.

Electrostatic is the branch of science that deals with the phenomena associated with electricity at rest.

It has been already discussed that generally an atom is electrically neutral that is in a normal atom the aggregate of positive charge of proton is exactly equal to the aggregate of negative charge of the electrons.

If some electrons are removed from the body of an atom, then it will left with a positive charge.

Then the atom is said to be positively charged. Whereas if some electrons are added to it, negative charge out balance the positive charge and the body is said to be negatively charged. We can therefore say that positive electrification of a body result from a deficiency of electrons whereas negative electrification result from an excess of electrons.

The total deficiency or excess of electrons in a body is known as its charge.

TYPES OF CHARGES

There are two types of charges, namely:

- (i) Positive charge
- (ii) Negative charge

WAYS OF PRODUCING CHARGES

Basically charges can be produced in the following ways or methods;

- (i) By friction
- (ii) By contact
- (iii) By electrostatic induction.

LAWS OF ELECTROSTATIC

(i) First law of electrostatics, states that like charges repel each other, whereas unlike charges attract each other. This law is the fundamental of static electricity.

(ii) Second law (Coulomb's law); states that the electric force between two point charges, q_1 and q_2 separated by a distance r is directly proportional to the product of their charges and inversely proportional to the square of the distance between the charges.

Coulomb's law can be mathematically express as;

$$F \propto q_1 q_2 / r^2 \quad \dots\dots\dots (12.a)$$

The magnitude of this force is given by;

$$F = K q_1 q_2 / r^2 \quad \dots\dots\dots (12.b)$$

Where K is a constant of proportionality.

F is in Newton's (N),

q_1 and q_2 are in coulombs (C) and

r is in meters.

The constant K is usually taken as;

$$K = 1 / 4\pi\epsilon \quad \dots\dots\dots (12.c)$$

$\epsilon = \epsilon_0 \epsilon_r$

Where ϵ_0 = absolute permittivity = $8.854 \times 10^{-12} \text{C}^2\text{N}^{-1}\text{m}^{-2}$ or $8.854 \times 10^{-12} \text{Fm}^{-1}$.

ϵ_r = relative permittivity = 1 (in air or vacuum).

Therefore $K = 1 / 4\pi\epsilon = 9.0 \times 10^9 \text{Nm}^2\text{C}^{-2} \quad \dots\dots\dots (12.d)$

Substituting the value of K in equation (12.b) above we get

$$F = \frac{1}{4\pi\epsilon_0} \times \frac{Qq}{r^2}$$

$$R = E = \frac{F}{q}$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{Qq}{r^2} \times \frac{1}{q}$$

$$= \frac{Q}{4\pi\epsilon_0} = \frac{Qq}{r^2}$$

or

$$\frac{1}{4\pi\epsilon_0} \times \frac{Qq^2}{r^2}$$

$$\frac{Qq^2}{4\pi\epsilon_0 r^2}$$

$$\frac{1}{4\pi\epsilon_0} \times \frac{Qq}{r^2}$$

$$\frac{1}{4\pi\epsilon_0} \times \frac{Q}{r}$$

$$\frac{Q}{4\pi\epsilon_0 r^2}$$

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$

$$F = 9.0 \times 10^9 \times q_1 q_2 / r^2 \dots\dots\dots (12.e)$$

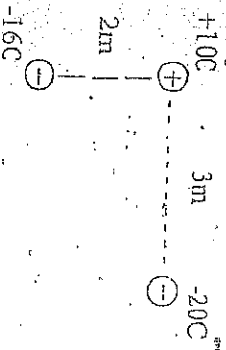
Example (12): Find the force of repulsion between two protons of 1.6×10^{-19} C each, if the distance between them is 1.0 nm.

Solution : since $q_1 = q_2 = q$

$$F = 9.0 \times 10^9 \times q^2 / r^2$$

$$F = 9.0 \times 10^9 \times (1.6 \times 10^{-19})^2 / (0.001)^2 = 2.304 \times 10^{-28} / 0.0001 = 2.3 \times 10^{-24} \text{ N.}$$

Example (13): three charges +10C, -20C, -16C are distributed as shown below. Find the resultant force acting on the 10C charge.



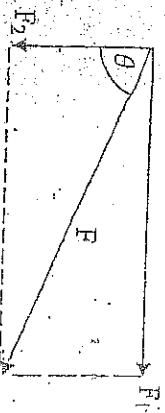
Solution:

Force of attraction between +10C and -20C is

$$F_1 = 9.0 \times 10^9 \times 10 \times 20 / (3)^2 = 2 \times 10^{11} \text{ N.}$$

Force of attraction between +10C and -16C is

$$F_2 = 9.0 \times 10^9 \times 10 \times 16 / (5)^2 = 3.6 \times 10^{11} \text{ N.}$$



$$\text{Resultant force } F = \sqrt{(F_1)^2 + (F_2)^2} = \sqrt{(2 \times 10^{11})^2 + (3.6 \times 10^{11})^2} = 4.12 \times 10^{11} \text{ N}$$

$$F = 4.12 \times 10^{11} \text{ N}$$

$$\tan \theta = F_1 / F_2 = 2 \times 10^{11} / 3.6 \times 10^{11} = 0.5555.$$

$$\theta = \tan^{-1} 0.55 = 29^\circ$$

Net force is 4.12×10^{11} N in a direction 29° to the line joining the +10C and -16C charge.

FIELD STRENGTH OR FIELD INTENSITY OR ELECTRIC INTENSITY (E).

Electric intensity at any point within an electric field may be defined as the force per unit positive charge (q) at that point. Or defined as the force experienced by a unit positive charge placed at that point. Its direction is the direction along which the force acts.

$$E = F / q \dots\dots\dots \text{N/C} \dots\dots\dots (13.a)$$

If we assume a point charge Q and a test charge q separated by a distance r in a vacuum,



The force between point charge Q and test charge q situated at point B is

$$F = 1 / 4\pi\epsilon_0 \times Q q / r^2$$

$$\text{Field intensity at B is } E = F / q = 1 / 4\pi\epsilon_0 \times Q q / r^2 \dots\dots\dots (13.b)$$

$$E = 1 / 4\pi\epsilon_0 \times Q q / r^2 q = Q / 4\pi\epsilon_0 r^2$$

Therefore $E = Q / 4\pi\epsilon_0 r^2$ (13.c)

ELECTRIC FLUX DENSITY OR ELECTRIC DISPLACEMENT:

Electric flux density is the number flux per unit area. If a flux of 4 coulombs passes normally through an area of $4\pi r^2$, then flux density is

$$D = 4 / 4\pi \text{ C/m}^2$$

It's related to electric field intensity by the relation,

$$D = \epsilon_0 \epsilon_r E \quad (\text{in a medium})$$

$$D = \epsilon_0 E \quad (\text{in free space vacuum})$$

Let us find the value of D at a point distance r meters from a point charge of Q coulombs. Imagine a sphere of radius r meters surrounding the charge. Total flux = Q coulombs and it falls normally on a surface area of $4\pi r^2$ meters.

$$\text{Hence, electric flux density } D = Q / 4\pi r^2 \text{ C/m}^2$$

ELECTRIC POTENTIAL (V):

Electric potential at a point is defined as the work done in bringing unit positive charge from infinity to that point against the electrical force of the field. The unit of potential will depend on the unit charge taken and the work done.

If in shifting one coulomb from infinity to a certain point in the electric field, the work done is one joule then potential of that point is one volt.

Obviously potential is work per unit charge.

$$\text{Therefore } 1 \text{ volt} = 1 \text{ joules} / 1 \text{ coulomb.}$$

Then electric potential;

$$V = W / q.$$

Electric potential is related to field intensity by the relation;

$$E = V / d \quad \text{V/m}$$

$$V = Ed \quad \text{Nc}^{-1}\text{m.}$$

$$\text{From the value of } E = q / 4\pi\epsilon_0 d^2 = q / 4\pi\epsilon_0 r^2$$

Where $d = r =$ distance between point charges.

$$\text{Therefore } V = q / 4\pi\epsilon_0 d^2 \times d = q / 4\pi\epsilon_0 d = q / 4\pi\epsilon_0 r$$

ELECTRIC FLUX:

The number of faraday tube of flux passing through a surface in an electric field is called the electric flux (or dielectric flux) through that surface. Electric flux is represented by the symbol Ψ , since electric flux is numerically equal to the charge it's measured in coulombs.

$$\text{Hence } \Psi = Q \text{ coulombs.}$$

Example (14): a point charge of 10^{-9} C is placed at a point A in free space. Calculate

- The intensity of electrostatic field on the surface of sphere of radius 5 cm and center A.
- The difference of potential between two points 20 cm and 10 cm away from the charge A.

$$\text{Solution: (i) using } E = q / 4\pi\epsilon_0 r^2 = 9.0 \times 10^{-9} \times 10^{-9} / (5 \times 10^{-2})^2 = 3600 \text{ V/m.} \Rightarrow \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2}$$

$$E = 3600 \text{ V/m}$$

- Potential at first point $r = 20$ cm

$$V_1 = q / 4\pi\epsilon_0 r = 9.0 \times 10^{-9} \times 10^{-9} / 20 \times 10^{-2} = 45 \text{ V.} \Rightarrow \frac{1}{4\pi\epsilon_0} \times \frac{q}{r}$$

$$V_1 = 45 \text{ V.}$$

$$\text{Potential at second point } r = 10 \text{ cm}$$

$$V_2 = 9.0 \times 10^{-9} \times 10^{-9} / 10 \times 10^{-2} = 90 \text{ V.}$$

$$V_2 = 90 \text{ V.}$$

Therefore the potential difference between the two points = $V_2 - V_1 = 90 \text{ V} - 45 \text{ V} = 45 \text{ V}$

$$p.d = 45 \text{ V}$$

Example (15): A parallel plate capacitor has plates 0.15 mm apart and dielectric with relative permittivity

of 3. Find the electric field intensity and the voltage between the plate, if the surface charge is $5 \times 10^{-4} \text{ C/cm}^2$.

Solution: electric intensity between plates is

$$E = \frac{\rho_s}{\epsilon_0 \epsilon_r} \text{ V/m}$$

$$\text{Charge density} = 5 \times 10^{-4} \text{ NC/cm}^2 = \text{Flux density} = D$$

$$\text{Therefore } D = 5 \times 10^{-6} \text{ C/m}^2$$

$$E = \frac{1}{\epsilon_0} \epsilon_r = 5 \times 10^{-6} / 8.854 \times 10^{-12} \times 3 = 188000 \text{ V/m.}$$

$$E = 188 \text{ KV/m.}$$

$$\text{Now potential difference } V = E \times r = 188000 \times (0.15 \times 10^{-3}) = 28.2 \text{ V.}$$

$$\therefore \text{Therefore } V = 28.2 \text{ V}$$

EXERCISE (3)

- (1) How much work is required to carry a charge of $5 \times 10^{-4} \text{ C}$ from a point 75 cm from a charge of $4 \times 10^{-3} \text{ C}$ to a point 35 cm from it?
- (2) Three point charges in air are located as follows;
 $+4 \times 10^{-8} \text{ C}$ at (0,0) meters, $-6 \times 10^{-8} \text{ C}$ at (4,0) meters and $+5 \times 10^{-8} \text{ C}$ at (0,3) meters. Find the electric field intensity at (4, 3) meters.

CAPACITANCE

Capacitor, is a device for storing electric charges or for storing electricity. It consists essentially of two metal plates carrying opposite charges. The metal plates are separated a small distance d by an insulator.

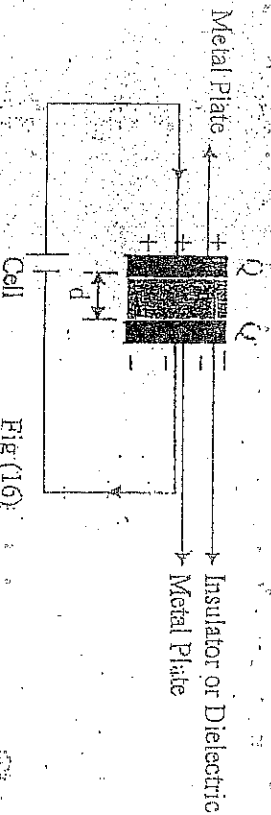


Fig (16)

The property of a capacitor to 'store electricity' may be called its capacitance. The capacitance of a capacitor is defined as the amount of charge required to create a unit p.d between its plates.

Suppose, we give Q coulomb of charge to one of the two plate of a capacitor and if a p.d of V volts is established between the two, then its capacitance is

$$C = Q/V = \text{charge} / \text{potential difference.}$$

Hence capacitance is the charge required per unit potential difference. By definition the unit of capacitance is coulomb per volt, which is also called farad (in honor of Michael faraday)

$$1 \text{ farad is equal to } 1 \text{ coulomb per volt.}$$

One farad is defined as the capacitance of a capacitor which requires a charge of one coulomb to establish a p.d one volt between its plates.

PARALLEL-PLATE CAPACITOR

(i) Uniform dielectric medium.

A parallel plate capacitor consisting of two plates M and N each of area $A \text{ m}^2$ separated by a thickness of d meters of a medium of relative permittivity ϵ_r as shown below.

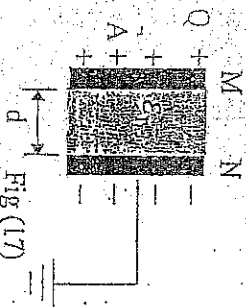


Fig (17)

If a charge of $+Q$ coulomb is given to plate M then flux passing through the medium is $\psi = Q$ coulomb. Flux density in the medium is. $D = \psi / A = Q / A$.

Electric intensity $E = V/d$ and $D = \epsilon E$

Therefore $Q/A = \epsilon V/d$ also $(Q/V = \epsilon A/d)$

Therefore $C = \epsilon A/d = \epsilon_0 \epsilon_r A/d$ Farad (in a medium)

Also $C = \epsilon_0 A/d$ Farad (in air or vacuum)

(ii) Composite medium:

Capacitor plates, and V_1, V_2, V_3 , the potential difference across the three slabs as shown below,

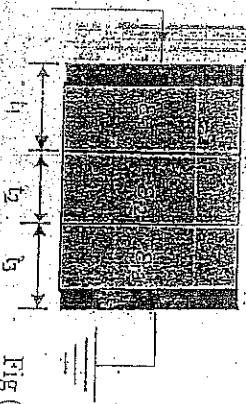


Fig (18)

Then, $V = V_1 + V_2 + V_3 = E_1 h_1 + E_2 h_2 + E_3 h_3$

Where h_1, h_2, h_3 are the thickness of the slabs

$$V = D / \epsilon_0 \epsilon_1 \times h_1 + D / \epsilon_0 \epsilon_2 \times h_2 + D / \epsilon_0 \epsilon_3 \times h_3$$

$$= D / \epsilon_0 (h_1 / \epsilon_1 + h_2 / \epsilon_2 + h_3 / \epsilon_3)$$

$$= Q / \epsilon_0 A (h_1 / \epsilon_1 + h_2 / \epsilon_2 + h_3 / \epsilon_3)$$

Therefore $C = Q/V$

$$C = \epsilon_0 A / (h_1 / \epsilon_1 + h_2 / \epsilon_2 + h_3 / \epsilon_3)$$

CAPACITORS IN SERIES

Consider three capacitors C_1, C_2 and C_3 are connected in series with potential difference of volts V applied across them as shown below.

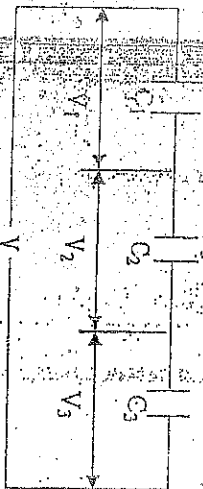


Fig (19)

In series combination charge on all capacitors is the same but p.d across each is different.

Therefore, $V = V_1 + V_2 + V_3$ or

$$Q/C = Q_1/C_1 + Q_2/C_2 + Q_3/C_3 \text{ or}$$

$$1/C = 1/C_1 + 1/C_2 + 1/C_3$$

For a charging applied voltage

$$dv/dt = dv_1/dt + dv_2/dt + dv_3/dt$$

The values of V_1, V_2 and V_3 can be found in terms of V .

$$\text{Now } Q = C_1 V_1 = C_2 V_2 = C_3 V_3 = CV$$

$$\text{Where } C = C_1 C_2 C_3 / C_1 C_2 + C_2 C_3 + C_3 C_1$$

$$\text{Therefore } C_1 V_1 = CV \text{ or } V_1 = VC/C_1$$

Similarly,

$$V_1 = V \times C_2 C_3 / C_1 C_2 + C_2 C_3 + C_1 C_3.$$

$$V_2 = V \times C_1 C_3 / C_1 C_2 + C_2 C_3 + C_1 C_3.$$

$$V_3 = V \times C_1 C_2 / C_1 C_2 + C_2 C_3 + C_1 C_3.$$

CAPACITORS IN PARALLEL.

Consider three capacitor are connected in parallel with potential difference of V volts V applied across them as shown below.

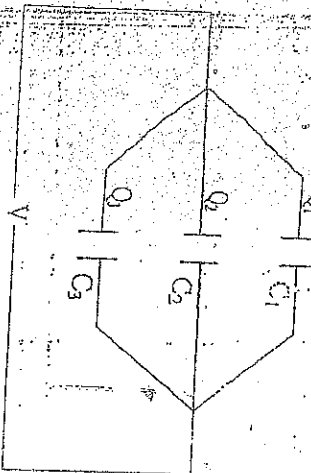


Fig (20)

In parallel connections, the p.d across each capacitor is the same but charge on each is different. Therefore $Q = Q_1 + Q_2 + Q_3$ or $CV = C_1 V + C_2 V + C_3 V$.
 $C = C_1 + C_2 + C_3$.

For such combination dW/dt is the same for all capacitors.

ENERGY STORED IN A CAPACITOR.

Charging of a capacitor always involves some expenditure of energy by the charging agency. Let us find the energy spend in charging a capacitor of capacitance C to a voltage V .

Suppose at any stage of charging the p.d across the plates is V .

By definition, energy store in a capacitor is equal to the work done in shifting one coulomb from one plates to another. If dq is charge next transferred the work done is

$$dW = V dq.$$

$$\text{Now } q = CV.$$

$$dq = C dV$$

Then, we have, $dW = Cvdv$.

Total work done in giving V units of potential is

$$\int dW = \int_0^V Cvdv.$$

$$W = \int_0^V Cvdv = C \left[\frac{V^2}{2} \right]_0^V = \frac{1}{2} CV^2.$$

If C is in farads and V is in volts, then $W = \frac{1}{2} CV^2$ joules.

Also

$W = \frac{1}{2} QV$ joules. $= Q^2 / 2C$ joules.

Therefore as seen from above, energy stored in a capacitor is

$$E = \frac{1}{2} CV^2$$

Now for a capacitor of plate area A m² and dielectric of thickness d meters energy per unit volume of dielectric medium,

$$E = \frac{1}{2} \frac{CV^2}{Ad} = \frac{1}{2} \frac{\epsilon A \times d \times V^2}{Ad} = \frac{1}{2} \epsilon \left(\frac{V}{d} \right)^2 = \frac{1}{2} \epsilon E^2 \text{ joules.}$$

Example (16); the capacitance of a capacitor formed by two parallel metal plates each 200 cm² in area,

- The total charge on the plates
- The electric potential in V/m .